

From Silicon to Rack: Arm and Meta Scaling AI Infrastructure through OCP

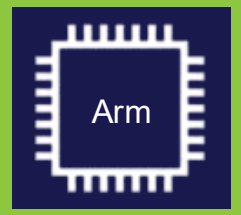
Prasanna Jayaraman, Solution Architect, Arm

Damien Chong, Hardware Engineer, Meta

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Scaling AI Infrastructure

Beyond Silicon Performance



- AI infrastructure evolving from individual components to tightly engineered systems
- Performance, power, and efficiency increasingly determined at the system level

Scaling from Silicon To Rack



- Compute density, power delivery, cooling, and integration at rack scale
- Coordinated design across silicon, server, and rack

Open Standards Enable Scale



- Interoperable designs across the infrastructure stack
- Reference architectures reduce integration complexity
- Faster ecosystem adoption and deployment

arm AGI CPU

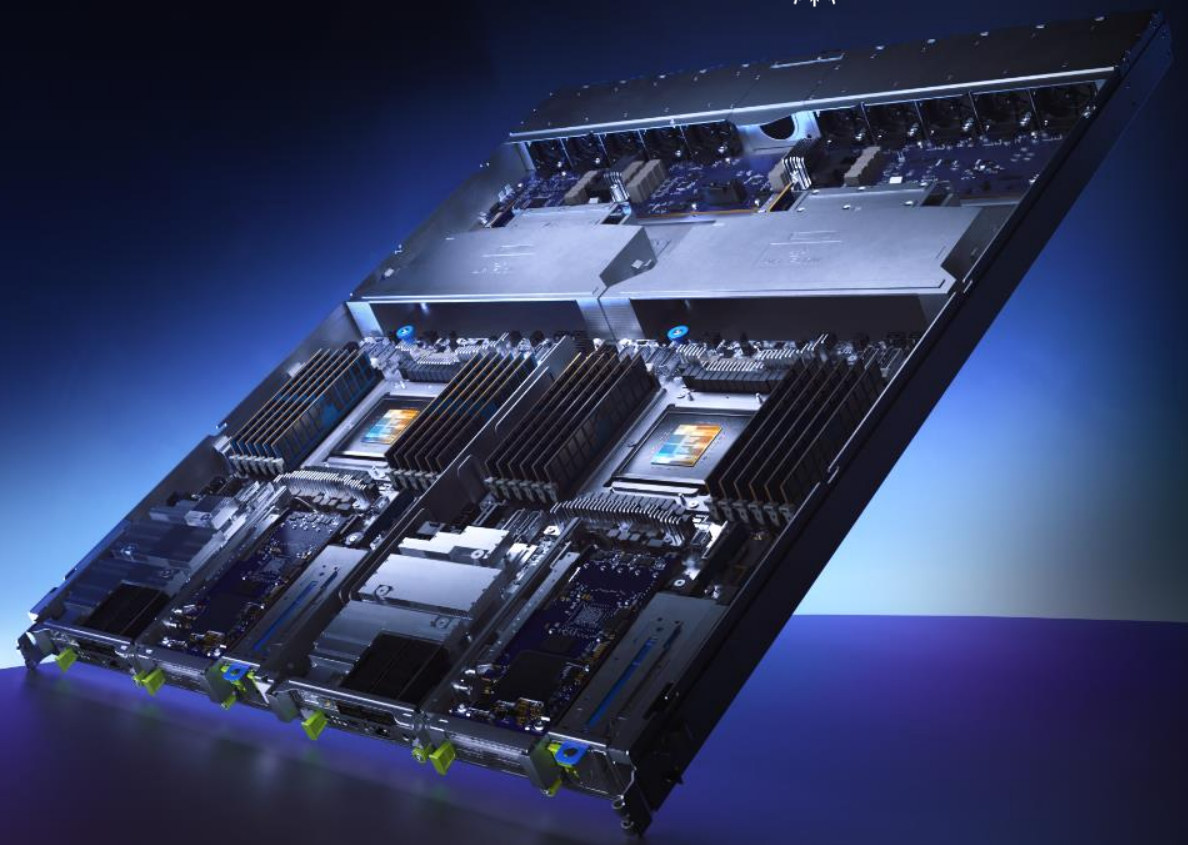
Arm AGI CPU 1OU Dual Node
Reference Server

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Open by design.
Scaled by the ecosystem.

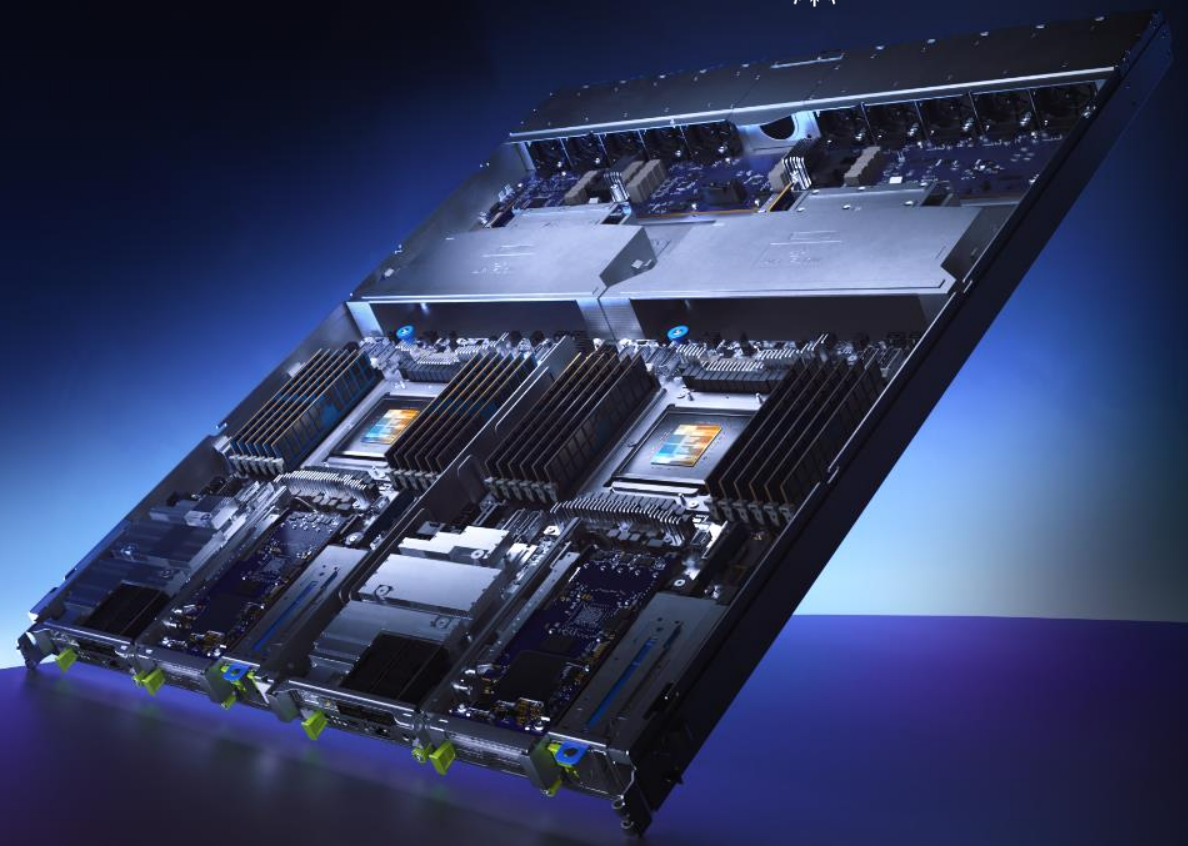
Open Rack V3

- Data Center Secure Control Module 2.0
- Data Center Modular Hardware System 1.1 RC2
- Foundation Chiplet System Architecture Alignment



arm AGI CPU

Arm AGI CPU 1OU Dual Node Reference Server



Accelerate Deployment

Provide reference design to shorten TTM

High Performance Compute

136 Neoverse V3AE cores
12 DDR5 DIMMs up to 8400 MT/s
96 lanes PCIe Gen6, CXL 3.0

Open Standards Alignment

ORv3 and DC-MHS Alignment;
OCP DC-MHS & DC-SCM 2.1
OCP NIC 3.0
OCP Open Platform Firmware (OPF) with EDK2 and OpenBMC
OCP HW management specifications
OCP FCSA
SBSA, SBBR/SBMR

Reusable Design Collateral(available under NDA):

Design and User Guide, HPM schematic and layout, stack-up, Routing Guidelines, Reference BIOS and BMC FW, BOM

arm AGI CPU

Arm AGI CPU 1OU Dual Node Reference Server

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OCP-compliant

Open 1OU dual-node reference server design compliant to OCP Standards

Density gain

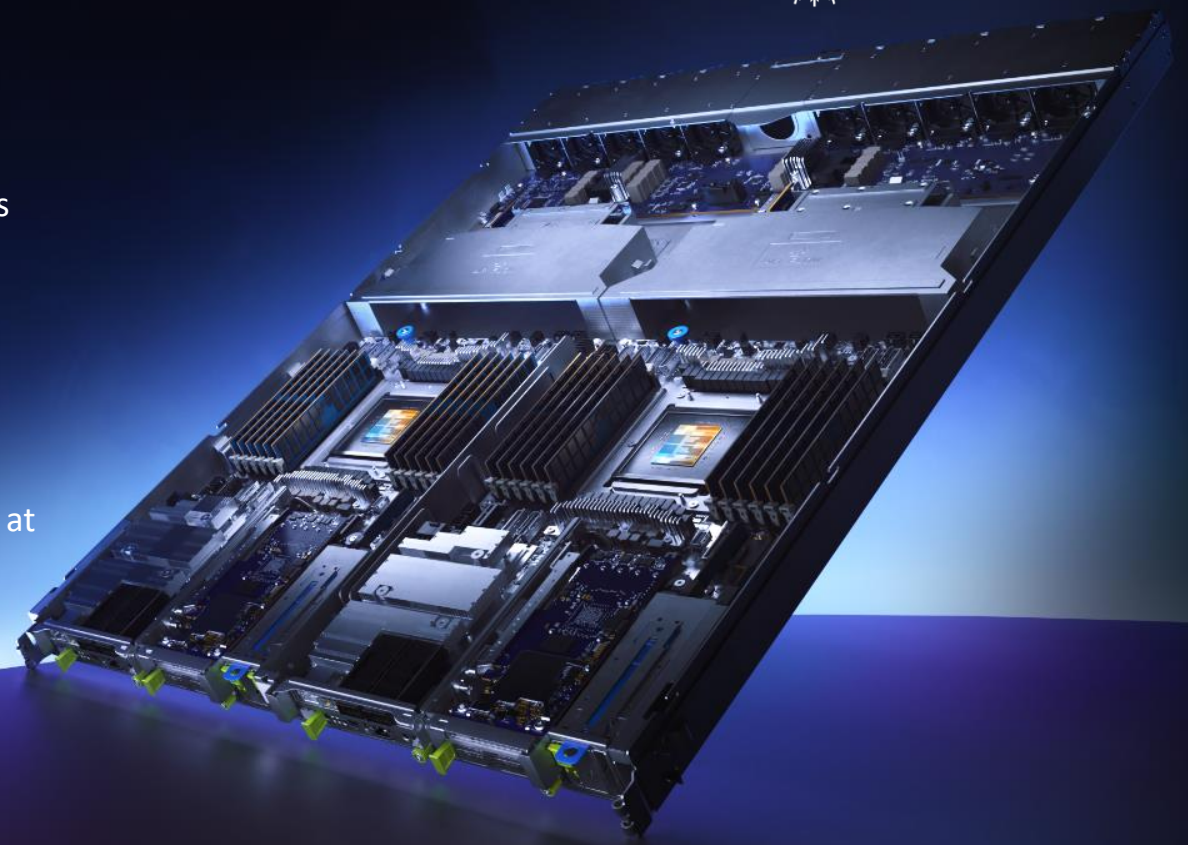
High-performance dual-node compute packed into just 1OU
Air-cooled designs leading to improved TCO for AI datacenters

Designed for rack-scale

Optimized for tightly-integrated rack level solution giving 2x performance at scale

Chiplet-aligned

Architecture follows FCSA-aligned chiplet design principles and standard interfaces for modularity and ecosystem interoperability



One Reference Architecture. Two Platforms. Minimal Redesign

Demonstrating design reuse and adaptability across different server use cases

Minimal Design Modifications. Multiple Meta Platform Implementations

Accelerating development through reference design reuse



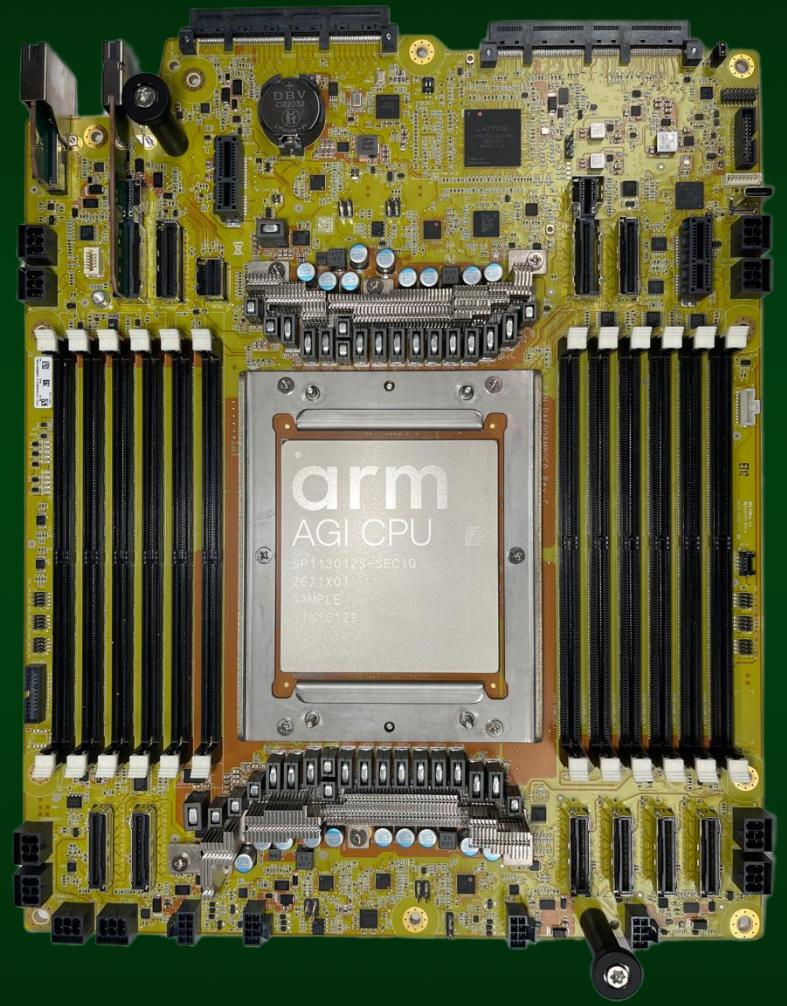
Arm AGI CPU in Meta Yosemite V5 Platform

- 1OU Half Width Server
  - 12ch DDR5 8000
  - 1x E1.S 9.5mm
  - 1x E1.S 25mm
- 6 servers in chassis
- OCP Technologies:
  - OCP NIC, DC-SCM & ORv3



Arm AGI CPU in Meta Rainier HPM

- DC-MHS SDNO B305 compatible
- OCP NIC & DC-SCM
- PICPWR 2x3
- 2x E1.S in Innovation Area
- Contribution to OCP after PVT



Intended Alignment with OCP Tenets

Openness

Intended to support open infrastructure

OCP-aligned mechanical, power, management, and expansion interfaces

ORv3 rack compatibility; DC-MHS-style motherboard design; DC-SCM-based management

Standard PCIe/CXL expansion

Efficiency

Designed to improve compute efficiency

High core density and high memory bandwidth

PCIe Gen6 expansion, CXL capability, and platform-level power and thermal management

All of this fits in a 1OU chassis giving significant TCO improvements

Impact

Lowers the barrier to Arm platform development and adoption

Reference system for software enablement, workload evaluation & PoCs

Design collateral helps shorten platform development and validation

Scale

Reusable architecture for cloud and AI infrastructure deployments

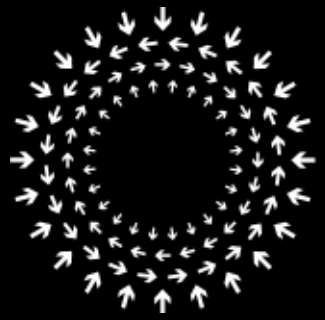
Standardized building blocks enable OEM/ODM implementation

Common HW, FW & management architecture supports repeatable platform designs

Sustainability

Intended to support energy-efficient, high-density compute (power/thermal management, air-cooled)

Reuse of open building blocks and design collateral reduces duplicated engineering and validation



OPEN Compute Project®

+

arm

Arm contributions across the full stack

Chiplet enablement

Server standardization

Server designs

Diverse silicon ecosystem

- Foundation Chiplet System Architecture (FCSA)
- Modular integration of compute, memory, and I/O
- Reduced design complexity for custom silicon

Software just works on Arm servers

- Standardized server architecture (SBSA, SBMR)*
- Compliance and validation framework (ADAC)
- Built-in system management and security

** Contributions in progress*

Helping OEMs launch Arm servers faster

- Pre-defined server hardware specifications^
- Aligned with Meta Yosemite V5 rack design
- Firmware development frameworks

^ Future contribution

arm

2X performance at scale*.
Efficiency by design.
Built for agentic.

*

Visit us at booth C72



Questions



Thank You!

